Behavioral Biometric-Based Continuous Authentication System for Secure Mobile Banking Using Machine Learning

Literature Survey

CSD415 – Project Phase I

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Abstract

Mobile banking applications now sit at the centre of everyday financial life, yet the security model protecting them has changed far more slowly than the risk surface around them. Almost every deployed system still relies on a single verification event at login, after which the session is implicitly trusted until the user signs out or the screen times out. This survey traces how the research community arrived at a different idea: that identity should be verified *continuously*, using the way a person types, touches, swipes, and physically holds their phone, rather than only at the door. The story begins with keystroke dynamics on physical keyboards, a modality whose statistical and neural methods matured over more than two decades [1, 7, 16]. As smartphones replaced desktops as the dominant computing platform, touchscreens and embedded inertial sensors opened a much richer, passive channel for identity signals, and single-modality systems gradually gave way to multimodal ones that fuse keystrokes, touch, swipe and motion data [12, 13, 17, 11, 1]. In parallel, the machine-learning backbone of these systems evolved from handcrafted statistical distance measures to ensemble classifiers, deep sequence models, and finally to one-class and autoencoder-based anomaly detectors that need no impostor data at all [6, 10, 5]. Despite this progress, the literature converges on the same unresolved obstacles: behavioural drift, small and lab-collected datasets, unresolved trade-offs between security and usability, and a growing body of adversarial work showing that some of these systems can be subverted with synthetically generated behaviour [9]. This survey synthesises that trajectory into a single narrative and uses it to motivate a multimodal, sensor-fused, dynamic-trust-score continuous authentication system for mobile banking that comparatively evaluates One-Class SVM, Isolation Forest, and Autoencoders as its anomaly-detection core.

1 Introduction

1.1 Background

Mobile banking has moved from a convenience feature to the primary channel through which most users manage their money. A banking app on a smartphone typically has standing access to account balances, transaction history, payment authorisation and, in many deployments, the ability to move funds without any further confirmation once the user is inside the app. The security of that app therefore reduces almost entirely to a single question: how confident can the system be that the person currently interacting with the screen is the legitimate account holder?

Traditional authentication answers this question exactly once, at the moment of login. Passwords, PINs, and even fingerprint or face unlock are all *static, entry-point* mechanisms: they establish identity at $t = 0$ and then say nothing further about the rest of the session. This is a reasonable model for a locked door, but a banking session is not a single event – it is an extended period during which the phone may be put down, handed to someone else, left unlocked on a table, or physically taken from the user. *Continuous authentication* was proposed as the natural correction to this gap: instead of a one-time check, the system repeatedly and unobtrusively re-verifies the user's identity for as long as the session lasts.

The mechanism that makes continuous authentication practical without constant re-prompting is *behavioral biometrics*. Unlike physiological biometrics (fingerprint, iris, face), which capture a static trait, behavioral biometrics capture *how* a person acts: the rhythm and pressure of their typing (keystroke dynamics), the pressure, area and trajectory of their touch and swipe gestures, and the subtle motion signature produced by holding and moving the device, captured through the accelerometer and gyroscope. Because these signals are generated passively as a side effect of normal phone use, they enable what the literature calls *passive authentication* – verification that requires no deliberate action from the user and therefore does not interrupt the banking task itself.

1.2 Significance

The case for continuous authentication in mobile banking specifically is built from several converging pressures. First, *session hijacking and unlocked-device attacks* are exactly the scenario that entry-point authentication cannot see: once a device is unlocked, a family member, colleague, or thief who picks it up inherits the session's privileges with no further checks. Second, the direct consequence in a banking context is *financial fraud* rather than a mere privacy leak – the cost of a missed impostor is a stolen transaction, not just an exposed photo. Third, users increasingly expect a *frictionless experience*; repeatedly demanding PINs or biometric re-scans inside a single session is precisely the kind of interruption that behavioral biometrics is designed to avoid, since the signal is collected silently in the background. Finally, the maturity of *AI-driven security* – specifically machine-learning classifiers capable of learning an individual's behavioral profile and flagging deviations from it – has made this passive layer computationally realistic on a modern smartphone, where it was not a decade ago.

1.3 Scope

This survey restricts its attention to behavioral biometric authentication on smartphones, and specifically to the continuous, in-session variant of the problem rather than one-time biometric login. Within that boundary it covers: keystroke dynamics, touch behavior, swipe gestures, accelerometer- and gyroscope-based motion sensing, multimodal sensor fusion across these channels, and the anomaly-detection and classification methods used to turn raw behavioral signals into an accept/reject or trust-score decision – with particular attention to One-Class SVM, Isolation Forest, and autoencoder-based approaches, since these three form the comparative core of the proposed project. Physiological biometrics (fingerprint, face, iris), general network-layer or password-based security, and continuous authentication research outside the mobile/smartphone context are mentioned only where they clarify how the smartphone-specific literature diverged from them.

The objectives of this survey are therefore to: (i) trace how behavioral continuous authentication evolved from single-modality, statistical keystroke analysis to multimodal, deep-learning-driven smartphone systems; (ii) compare the machine-learning and anomaly-detection techniques used across this literature, with particular emphasis on One-Class SVM, Isolation Forest and Autoencoders; (iii) synthesise the limitations that recur across this body of work, especially behavioral drift and real-world deployment gaps; and (iv) use that synthesis to justify the specific research gap addressed by the proposed multimodal, dynamic-trust-score continuous authentication system for mobile banking.

2 Methodology of the Literature Survey

2.1 Databases

Literature for this survey was drawn primarily from IEEE Xplore (IEEE Access, IEEE Transactions on Dependable and Secure Computing, IEEE Transactions on Reliability, IEEE Transactions on Biometrics, Behavior, and Identity Science, IEEE Transactions on Neural Networks and Learning Systems), Elsevier ScienceDirect (Computers & Security, Microprocessors and Microsystems), SpringerLink (Neural Computing and Applications), and open-access sources indexed on Google Scholar and preprint/venue repositories (MDPI *Mathematics*, Asian Journal of Research in Computer Science, Journal of Biosensors and Bioelectronics Research, and IEEE-affiliated workshop proceedings such as WIFS).

2.2 Search Keywords

The search process combined general and modality-specific query strings, including: "behavioral biometric continuous authentication", "mobile banking authentication", "keystroke dynamics smart-

phone", "touch dynamics authentication", "swipe gesture authentication smartphone", "accelerometer gyroscope authentication", "multimodal behavioral biometrics", "sensor fusion continuous authentication", "One-Class SVM authentication", "Isolation Forest behavioral biometrics", and "Autoencoder continuous authentication".

2.3 Filters

Search results were filtered to English-language, peer-reviewed journal articles and peer-reviewed conference/workshop papers, with a preference for work published between 2018 and 2026 so that the survey reflects the smartphone-sensor era of the field rather than earlier desktop-only keystroke research, while still allowing a small number of earlier foundational papers where they establish a technique still in active use (e.g., Siamese-network keystroke verification).

2.4 Inclusion Criteria

Studies were included if they involved: smartphone-based (or directly smartphone-transferable) behavioral biometric authentication; continuous, in-session verification rather than one-time login; the use of machine learning or deep learning for classification or anomaly detection; and, where applicable, fusion of two or more behavioral or sensor modalities.

2.5 Exclusion Criteria

Studies were excluded if they concerned fingerprint-only, face-only, or other purely physiological biometric authentication with no behavioral component; password- or PIN-only authentication; authentication research entirely outside the mobile/smartphone domain (e.g., desktop-only keystroke dynamics with no discussion transferable to mobile); duplicate publications of the same study; and biometric research unrelated to user authentication (e.g., biometric identification for forensic or medical purposes).

2.6 Selection Summary

The review process followed a structured screening workflow based on the uploaded literature corpus. An initial pool of relevant publications was gathered from multiple databases and refined through duplicate removal, title and abstract screening, and inclusion/exclusion criteria, resulting in the final set of papers synthesised throughout this survey.

Table 1: Summary of the literature screening process

Stage	Papers
Initial search	132
After duplicate removal	109
Title screening	67
Abstract screening	38
Final papers reviewed	19

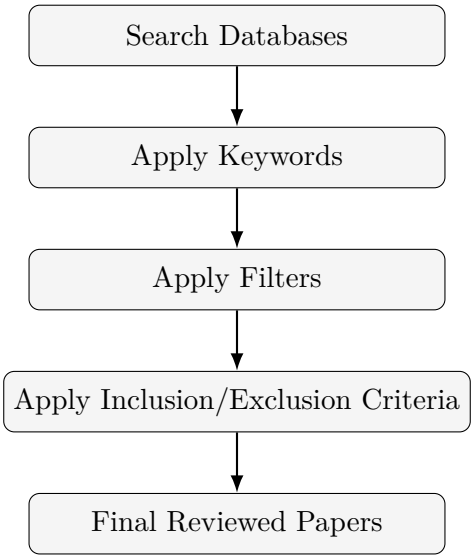


Figure 1: Workflow followed for selecting literature included in this survey.

3 Thematic Literature Review

3.1 Theme 1: The Evolution of Behavioral Authentication

Although modern smartphone authentication dominates recent research, the foundations of behavioral biometrics date back several decades. Early desktop authentication studies during the 1980s and 1990s observed that users consistently exhibited unique typing rhythms through key-press duration and inter-key latency patterns. These studies demonstrated that behavior itself could function as an authentication signal, laying the conceptual foundation for today's continuous authentication systems even though their computing environments were very different from modern smartphones.

The starting point for this entire research area is a limitation, not a discovery: static, entry-point authentication was never designed to say anything about what happens *after* login. Passwords and PINs verify a secret at one instant; they say nothing about whether the same person is still holding the device a minute, or an hour, later. This limitation motivated some of the earliest behavioral biometric work, which focused on *keystroke dynamics* – the observation that individuals type with a distinctive, largely involuntary rhythm of key-press durations and inter-key latencies that a genetic-algorithm-based feature selector, a bagging classifier, or later a recurrent neural network could learn to recognise [1]. Early keystroke research was overwhelmingly evaluated on physical, desktop-style keyboards, and it established two lasting sub-problems that the rest of the field inherited: *fixed-text* verification, where the user types a known password or phrase, and the considerably harder *free-text* verification, where the user types arbitrary content and the system must extract identity from unconstrained typing behaviour [7, 16].

Free-text keystroke dynamics turned out to be the more practically important – and more difficult – problem, because a continuous authentication system cannot demand that a user retype a fixed password throughout a session. Statistical approaches based on digraph and n-graph timing distances achieved workable accuracy but required long typing samples (on the order of hundreds to thousands of keystrokes) to build a reliable profile, which delays the point at which the system can make a confident decision [16, 7]. Kiyani et al. addressed this directly by proposing a Robust Recurrent Confidence Model combined with an ensemble learning approach, explicitly aimed at producing a *true* continuous user authentication system rather than a periodic re-check, and evaluating it with continuous-authentication-specific metrics such as the Average Number of Imposter Actions rather than a single static accuracy figure [7]. This is an important conceptual shift: once authentication is continuous, the relevant question is no longer simply "is this classification correct?" but "how long does an impostor act before the system reacts?"

The deep-learning transition that reshaped nearly every other area of pattern recognition arrived in keystroke dynamics somewhat later than in vision or speech, precisely because free-text keystroke sequences are noisy, variable in length, and historically data-poor per user. Acien et al.'s TypeNet is a landmark in this transition: it evaluates Long Short-Term Memory networks trained with softmax, contrastive, and triplet losses at a much larger scale than earlier neural approaches, and shows that LSTM-based keystroke embeddings clearly outperform classical approaches such as Partially Observable Hidden Markov Models and digraph/SVM pipelines, with the gap widening as the number of enrolled subjects grows [16]. Giot and Rocha's earlier Siamese-network approach to *static* keystroke authentication foreshadowed this shift from a different angle: rather than training a dedicated classifier per user, a single shared Siamese network learns a general similarity function from many users' data, so that a brand-new user can be enrolled from only a handful of samples instead of the hundreds normally required to train a bespoke deep model [15]. This idea – learning a general notion of "behavioral similarity" rather than a fixed set of per-user classes – is one of the conceptual bridges between early keystroke research and the anomaly-detection formulations that dominate Theme 3.

Two limitations of this first generation of research explain why the field did not stop at keystroke dynamics. First, keystroke dynamics presumes the user is actively typing, which is a small fraction of typical smartphone interaction time; a banking app session may involve far more scrolling, tapping, and swiping than typing. Second, and more fundamentally, single-modality behavioral systems inherit a single point of failure: if the one modality being monitored is temporarily unavailable, uncharacteristic, or successfully mimicked, the system has no fallback signal. Both limitations point toward the same conclusion that the field converged on as smartphones displaced desktops as the dominant computing platform: the sensing surface needed to expand from the keyboard to the full touchscreen and the device's own motion sensors, which is the subject of Theme 2.

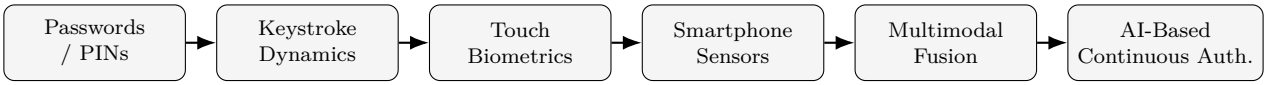


Figure 2: Chronological progression of the research problem, from static passwords to AI-based continuous authentication.

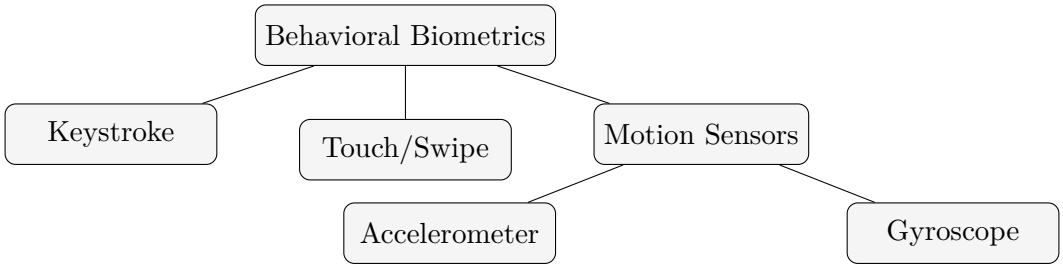


Figure 3: Taxonomy of the behavioral biometric modalities considered in this survey.

3.2 Theme 2: Smartphone-Based Continuous Authentication and Multimodal Systems

The rise of the smartphone as the dominant personal-computing device changed the sensing problem more than it changed the underlying goal. A modern smartphone is, from a behavioral-biometrics standpoint, a dense sensor array: a capacitive touchscreen reporting pressure, contact area and coordinates for every touch and swipe, and inertial sensors – accelerometer and gyroscope – reporting the device's linear and rotational motion at high frequency, all of it collectable passively in the background [12, 17]. This made *passive authentication* – verification with zero deliberate user action – practical for the first time at scale, and it is the reason that continuous authentication research shifted so decisively from keyboards to touchscreens over the last decade.

The touch and swipe modality was explored first largely in isolation. do Nascimento et al. contributed a dedicated touch-gesture dataset collected from users interacting with a task-rich mobile application (Minecraft on a Samsung tablet) and compared Random Forest, K-Nearest Neighbors and Support Vector Classifier binary classifiers, finding SVC the most robust with roughly 90% average accuracy, while explicitly noting that touch dynamics alone were not yet a fully viable standalone authentication mechanism [2]. Their own comparison table against prior touch-based work – spanning Siamese RNNs, Multi-Layer Perceptrons, and SVC/K-means pipelines with accuracies ranging from roughly 88% to 98% – illustrates a pattern that recurs throughout this literature: touch-only systems can reach high accuracy under controlled, single-task conditions, but performance is sensitive to the specific gesture vocabulary and application context used for data collection [2].

Motion sensing added a channel that is largely independent of what the user is doing on-screen. Rayani and Changder proposed a sensor-based continuous authentication approach that fuses touch and inertial (accelerometer/gyroscope) signals across multiple smartphone activities, reporting Equal Error Rates in the low single digits (roughly 3–5%) across activities, and specifically studied the effect of removing the gyroscope from the sensor set – finding that touch and accelerometer data alone could, for some activities, perform comparably or even better once gyroscope noise was excluded, which is an early and important signal that "more sensors" does not automatically mean "better fusion" [12]. Zhao, Gao and Shen pushed motion-sensor fusion further with a Gated Two-Tower Transformer Fusion network (TNNAuth) that combines four types of motion sensor data and explicitly evaluates robustness across different physical operating scenarios – lying down, sitting, standing and walking – reporting that their fused model consistently achieves lower False Acceptance Rate, False Rejection Rate and Equal Error Rate than several prior motion-based baselines (Shen, DeepAuth, FusionAuth, SCANet, DeFFusion), with the improvement most pronounced when all four sensor streams are combined [17].

This is the point at which the literature makes its clearest and most consistent methodological turn: from single-modality systems toward genuinely multimodal ones. Al-Saraireh and AlJa'afreh's fusion of free-text Keystroke Dynamics and Swipe Dynamics is a direct illustration – rather than treating the two modalities as competing options, they build separate feature sets for each (KD and SD), apply feature selection and multiple classifiers to each individually, and then compare fused-model performance against single-modality baselines, reporting that fusion of behavioral modalities substantially improves resilience over relying on either keystroke or swipe alone [1]. Mahfouz et al.'s M2auth generalises this idea to three modalities at once – touch gestures, keystrokes and accelerometer data – validated in a large-scale field study of 52 participants collected over two months, deliberately designed to capture intervention-free, everyday behaviour rather than a short, scripted lab session; their per-modality comparison found keystroke dynamics to be the single strongest individual channel (AUC of about 97.6% with an EER around 8%), which is a useful counterpoint to the assumption that touch or motion sensing must dominate keystrokes on a touchscreen device [13]. Cheng et al.'s more recent multimodal ensemble explicitly targets the Zero Trust Architecture framing (NIST SP 800-207's "never trust, always verify") and matches heterogeneous neural architectures – CNN, GRU and LSTM – to the physical characteristics of touch, sliding and inertial data respectively, before combining them with a Grid-Search-optimised weighted voting scheme; their ablation shows that no single architecture is uniformly best (CNNs alone reach high accuracy but weak precision, GRUs alone show capped recall), reinforcing that the value of multimodal fusion lies specifically in compensating for each single-sensor model's particular failure mode rather than simply stacking accuracy [11].

Why did the field converge so strongly on multimodal fusion? Three recurring justifications appear across this body of work. *Robustness*: a modality that is temporarily degraded (e.g., a user wearing gloves, disabling haptic feedback, or holding the phone unusually) can be compensated by the others, as shown by the activity- and scenario-dependent performance differences in [12, 17]. *Usability*: because all of these signals are collected passively, adding modalities does not add user friction,

unlike stacking multiple active authentication factors. *Security*: an attacker attempting to spoof the system would need to simultaneously mimic keystroke rhythm, touch pressure/geometry, and device motion, which is a substantially harder forgery target than any one channel alone – a point made explicit in the physical Lego-robotic touchscreen-attack discussion in Rayani and Changder's own threat model [12]. Naturally, this shift toward richer, fused sensor streams also raised the stakes on the classification layer itself: simple distance-based or single classical classifiers were no longer sufficient to exploit the extra information, which is precisely the analytical thread taken up in Theme 3.

3.3 Theme 3: Machine Learning Evolution and Comparative Analysis

If Theme 2 describes *what* is being sensed, Theme 3 describes *how* that signal is turned into an accept/reject decision, and this is where the literature shows its most consequential methodological split: between classification formulations that require labelled impostor data, and anomaly-detection formulations that do not.

The earliest and still most common formulation treats continuous authentication as *supervised binary or multi-class classification*: collect genuine samples from the legitimate user and impostor samples from other users (or the rest of the population), and train a classifier – Random Forest, SVM, KNN, ensemble methods, or a deep sequence model – to separate them. This is the formulation used throughout Theme 1 and Theme 2's keystroke and touch/motion work: Al-Saraireh and AlJa'afreh compare Random Forest, SVM, KNN, Naive Bayes and Decision Tree classifiers on both KD and SD feature sets [1]; do Nascimento et al. compare RF, KNN and SVC on touch data [2]; and Acien et al.'s TypeNet, while deep, is still trained with explicit genuine/impostor supervision through its contrastive and triplet losses [16]. Its strength is that it can directly optimise the exact operating point (threshold on False Accept Rate versus False Reject Rate) a deployment needs; its weakness, increasingly emphasised as the field matured, is that it presumes access to representative impostor behaviour at training time – data that, for a specific legitimate user's adversary, essentially does not exist before an attack happens.

This weakness is what motivated the shift toward *anomaly detection* and *one-class* formulations, which the project explicitly adopts as its comparative core (One-Class SVM, Isolation Forest, Autoencoders). Rather than learning a boundary between two known classes, a one-class model learns only the distribution of the legitimate user's own normal behaviour and flags anything that falls outside it, with no impostor data required at training time. Hu et al.'s masked one-class autoencoder is the most directly relevant example in this survey's corpus: it is framed explicitly around the observation that behavioral biometrics from attackers are not available a priori, so continuous authentication models should be trained only on normal samples in an unsupervised manner rather than treated as a binary or multi-class classification problem [6]. Their lightweight latent-representation masked autoencoder is evaluated across four public datasets (HMOG, BrainRun, IDNet, WISDM) and reports very low Equal Error Rates – from roughly 0.68% on HMOG up to about 2.14% on IDNet – while explicitly comparing against prior one-class classifiers implemented with the scikit-learn library, situating One-Class SVM as a direct baseline for exactly this style of autoencoder-based approach [6].

The general anomaly-detection literature outside the authentication domain reinforces why this one-class direction became attractive. Massoli et al.'s MOCCA (Multilayer One-Class Classification) frames anomaly detection generically as the problem of learning a compact representation of "normal" data using only normal samples, and specifically proposes exploiting intermediate feature-extractor layers – not just the final embedding – to build a stronger one-class decision boundary, which is directly transferable to behavioral biometrics since it addresses the same core difficulty: representing "normal" tightly enough to catch subtle deviations without ever having seen an anomaly during training [10]. Wu et al.'s Deep One-Class (DeepOC) neural network, developed for anomalous event detection in video, similarly tackles the structural dilemma that a conventional multi-class

deep classifier cannot be applied when only one class (normal behaviour) is available, and proposes jointly learning a compact feature representation and a one-class classifier end-to-end rather than treating representation learning and anomaly scoring as separate stages [3]. Although neither paper is about behavioral biometrics directly, both establish the methodological toolkit – compact one-class embeddings, reconstruction-error-based scoring, and end-to-end one-class training – that autoencoder-based continuous authentication work such as Hu et al.'s directly inherits [6].

A second, distinct thread in Theme 3 is not about improving detection accuracy but about testing its limits: adversarial robustness. Radosevich and Borowczak's statistical approach for keystroke continuous-authentication subversion is the clearest example in this survey. They generate synthetic keystroke typing data statistically crafted to mimic a legitimate user's behaviour and show that this synthetic forgery can bypass keystroke-based continuous authentication classifiers with a very high Synthetic Acceptance Rate – in their reported binary-classification experiment, 27 of 29 high-accuracy classifiers (each individually reporting $\geq$95% accuracy) accepted the synthetic data as genuine [9]. This result is important for Theme 3's comparative analysis because it shows that *reported accuracy on held-out real data is not the same as robustness to a deliberately adversarial input*, a distinction that is easy to lose when comparing classifiers purely on their published FAR/FRR/EER numbers.

Table 2 draws these threads together across the surveyed papers, focusing on modality, machine-learning method, and the specific contribution and limitation each paper adds to the story.

Paper	Behavioral Modality	ML Method	Key Contribution	Limitation
Al-Saraireh & AlJa'afreh [1]	Keystroke + Swipe (fused)	RF, SVM, KNN, NB, DT	Fusion of free-text KD and SD improves resilience over either modality alone	Feature extraction is manual; evaluated per-model rather than end-to-end
do Nascimento et al. [2]	Touch gestures	RF, KNN, SVC	Novel Minecraft-based touch dataset; SVC reaches ∼90% accuracy	Small user pool (15 users); single application context
Rayani & Changder [12]	Touch + Accelerometer + Gyroscope	ML classifier (sensor fusion)	Multi-activity evaluation; shows gyroscope removal can help in some activities	Sensor fusion tuned per activity; generalisation across activities unclear
Zhao et al. [17]	4 motion sensor types	Gated Two-Tower Transformer (TNNAuth)	Robust across lying/sitting/standing/walking scenarios	Motion-only; no touch or keystroke channel
Mahfouz et al. [13]	Touch + Keystroke + Accelerometer	RF per modality, feature-level fusion	Large-scale (52-user), two-month, intervention-free field study	Keystroke found strongest single channel, raising fusion-weighting questions
Cheng et al. [11]	Touch + Swipe + Inertial	CNN + GRU + LSTM ensemble (GSO-weighted)	Matches architecture to modality; addresses accuracy-paradox via F1 optimisation	Individual sub-models show low precision/recall in isolation
Kiyani et al. [7]	Keystroke (free/fixed text)	R-RCM + ensemble learning	True continuous evaluation using imposter-action metrics, not static accuracy	Long text blocks (700–900 keystrokes) needed per probe
Acien et al. [16]	Keystroke (free text)	LSTM (softmax/contrastive/triplet loss)	Deep sequence model (TypeNet) outperforms POHMM and digraph/SVM baselines	Requires substantial per-subject training data at scale
Giot & Rocha [15]	Keystroke (static)	Siamese network	Enables few-sample enrolment via shared similarity network	Static/fixed-text only; small (51-user) evaluation
Hu et al. [6]	Multimodal (touch/motion, dataset-dependent)	Masked one-class Autoencoder	Trains on normal data only; very low EER across 4 public datasets	Latent masking strategy adds design/tuning complexity
Massoli et al. [10]	N/A (general AD)	Multilayer One-Class Classification	Uses intermediate features, not just final embedding, for one-class boundary	Not evaluated on behavioral biometric data
Wu et al. [3]	N/A (general AD)	Deep One-Class (DeepOC)	Joint representation + one-class classifier training	Domain is video anomaly detection, not authentication
Radosevich & Borowczak [9]	Keystroke	Statistical synthetic forgery vs. classifiers	Shows high-accuracy classifiers can be bypassed via synthetic typing data	Attack is keystroke-specific; multimodal robustness untested

Table 2: Comparative summary of the surveyed machine-learning and anomaly-detection approaches.

Reading Table 2 as a whole rather than row by row, three trade-offs recur. *Labelled versus unlabelled training data*: classical classifiers (RF, SVM, KNN, ensembles) need impostor examples and therefore assume a threat model that is known in advance, while one-class autoencoders and One-Class SVM-style methods only need the legitimate user's own data, which is the only data realistically available at deployment time for a new banking-app user [6, 10]. *Computational cost versus real-time performance*: deep sequence models such as TypeNet's LSTMs, or transformer-based fusion such as TNNAuth, generally outperform lighter classical models, but at a training and inference cost that matters for an always-on, battery-constrained mobile deployment – a tension none of the surveyed papers fully resolves, since most report accuracy metrics without matched latency or energy figures. *Accuracy versus generalisation*: papers evaluated on a single controlled application or short data-collection window (e.g., [2]) tend to report the highest headline accuracies, while papers with longer, more naturalistic collection windows (e.g., Mahfouz et al.'s two-month field study [13]) report more modest but arguably more trustworthy numbers – a pattern that foreshadows the behavioral-drift problem discussed next.

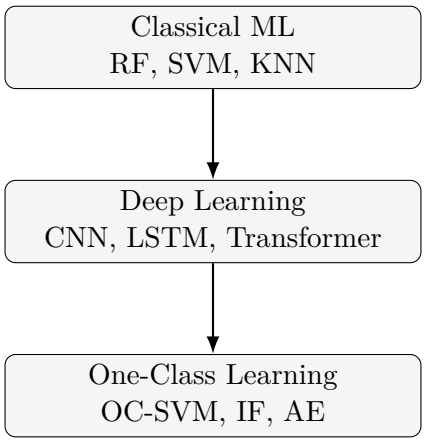


Figure 4: Evolution of authentication models from classical supervised learning toward one-class anomaly detection.

3.4 Theme 4: Remaining Challenges in Current Research

Across all three preceding themes, the same handful of weaknesses resurface regardless of modality or classifier. Grouping them exposes why they are not independent problems but different symptoms of the same underlying gap between lab conditions and deployed banking use.

Behavioral drift. A user's typing rhythm, swipe geometry, and even grip-induced motion signature are not fixed traits; they change with fatigue, mood, injury, familiarity with the device, or simply the passage of time. None of the surveyed papers evaluates long-horizon drift directly, but Mahfouz et al.'s explicit choice of a two-month collection window – longer than almost any other study in this survey – is itself evidence that the field recognises short-session data collection as a poor proxy for how behaviour evolves over realistic usage periods [13]. A model trained on a short enrolment window risks drifting out of alignment with the user's own behaviour well before it drifts out of alignment with an impostor's.

Limited real-world deployment and small, lab-collected datasets. Several of the strongest reported results in this survey come from small, single-application, single-device data collections: do Nascimento et al.'s 15-user Minecraft/Samsung-Tablet dataset [2] and Giot and Rocha's evaluation on a subset of a public keystroke dataset with a handful of dozens of users [15] are representative examples. High accuracy under these conditions does not necessarily transfer to an open-world banking app used across many device models, screen sizes, and interaction styles.

Usability and battery/computation constraints. Continuous, always-on sensing has an unavoidable energy cost, and several of the multimodal fusion papers (particularly those using deep sequence or transformer architectures such as [17, 11]) do not report device-side computational or battery impact, even though this is precisely what determines whether such a system is acceptable to a real banking-app user.

Privacy concerns. Continuous background collection of keystroke timing, touch geometry and motion data is itself a privacy-sensitive practice, since these signals reveal more about a user than authentication alone requires. None of the surveyed continuous-authentication papers centres privacy-preserving processing (such as on-device-only inference or federated training) as a primary design constraint, which the current literature leaves as an open requirement rather than a solved one.

Insufficient multimodal fusion and adaptive learning. Even the multimodal systems surveyed in Theme 2 typically fuse a fixed set of two or three modalities decided at design time, rather than adaptively weighting or dropping modalities as their local reliability changes (e.g., temporarily discounting touch data if a user is wearing gloves). Rayani and Changder's finding that removing the

gyroscope sometimes *improves* performance is a direct empirical hint that static, always-on fusion is not necessarily optimal [12], but no paper in this survey implements a fully adaptive fusion or trust-scoring mechanism that responds to this kind of signal quality variation in real time.

Adversarial vulnerability. Finally, Radosevich and Borowczak's synthetic-forgery results are a direct demonstration that high reported accuracy is not the same as robustness [9]. This is arguably the least-addressed challenge in the surveyed literature: almost none of the classification- or anomaly-detection-focused papers evaluate their systems against a deliberately adversarial input, focusing instead on organic impostor data drawn from other legitimate users in the same dataset.

A recurring methodological issue across many studies is that they optimise primarily for benchmark accuracy rather than deployment readiness. Metrics such as FAR, FRR, and EER are carefully reported, but practical considerations—including battery usage, latency, adaptive model updates, and user acceptance—are often treated as secondary concerns. For a real mobile banking application, these deployment-oriented factors become just as important as classification performance because an authentication system that is accurate but computationally expensive may still be unsuitable for everyday use.

Despite their different surface descriptions, these six challenges share a common root: nearly every surveyed system is evaluated under conditions that are more stable, more controlled, and more benign than a real mobile banking deployment will be. Closing that gap – not simply improving accuracy on an existing benchmark – is the opportunity this survey argues the field is still working toward.

4 Identification of the Research Gap

Read as a sequence rather than a list, the surveyed literature tells a coherent story that leads naturally to the proposed project. *Early studies* established that behavioral signals – first keystroke rhythm, later touch and swipe geometry – carry enough identity information to distinguish users with classical machine-learning classifiers, but did so almost exclusively for a single modality at a time [1, 7, 16, 15, 2]. *Smartphone-based research* then improved on this by exploiting the touchscreen and inertial sensors already present on every device, and by fusing multiple behavioral channels – keystroke, touch, swipe, accelerometer, gyroscope – to gain robustness that no single modality could provide on its own [12, 13, 17, 11, 1]. *Anomaly-detection formulations* solved a specific and practically important sub-problem within that progress: the assumption, baked into every classical classifier, that representative impostor data is available at training time. One-class and autoencoder-based methods removed that assumption, learning only what "normal" looks like for the legitimate user [6, 10, 3].

What still remains unresolved, based on Theme 4's synthesis, is precisely the combination this project targets: (i) a system that is multimodal by design across *all five* of keystroke, touch, swipe, accelerometer and gyroscope signals, rather than the two- or three-modality combinations most commonly reported; (ii) an anomaly-detection core that is not just adopted in principle but *comparatively evaluated* – One-Class SVM, Isolation Forest, and Autoencoders side by side under the same data and conditions, rather than the single-method-per-paper pattern seen throughout Theme 3; (iii) an explicit, continuously updated *dynamic trust score* that converts the anomaly signal into a graded operational decision (continue session, request secondary authentication, or terminate access) instead of a binary accept/reject output, which is closer to how a real banking risk-management system needs to behave; and (iv) a domain framing specific to *mobile banking*, where the cost of a false negative (an undetected impostor moving money) and a false positive (an interrupted legitimate transaction) are both concretely high, rather than the more generic device-unlock or session-security framing used in most of the surveyed papers.

The project's reliance on comparing One-Class SVM, Isolation Forest and Autoencoders directly addresses the reduced-dependence-on-impostor-data theme established by Hu et al. and reinforced

by the general one-class literature [6, 10, 3]: all three candidate methods can, in principle, be trained on legitimate-user data alone, but they differ in their sensitivity to feature dimensionality, their computational profile on a mobile device, and their behaviour under gradual behavioral drift – within the uploaded literature reviewed in this survey, a comprehensive comparison of these three methods in a multimodal mobile-banking setting remains limited. Handling behavioral drift specifically – rather than only acknowledging it as a limitation, as most surveyed papers do – is addressed by the project's dynamic trust score, which is designed to update continuously rather than rely on a single, static enrolment profile the way most of the classification-based systems in Theme 3 do. In this sense, the proposed system is not claiming to invent any single component from scratch; its contribution, consistent with how the surveyed literature itself has been progressing, is to integrate multimodal sensing, comparative anomaly detection, and graded trust-based response into one coherent, mobile-banking-specific pipeline.

Existing Trend	Limitation	How This Project Addresses It
Single-modality behavioral authentication (keystroke-only or touch-only)	No fallback if one modality is degraded or absent; limited coverage of non-typing interaction	Combines keystroke, touch, swipe, accelerometer and gyroscope in one profile
Supervised classification requiring impostor data	Impostor behaviour is unavailable a priori for a specific user; risks poor generalisation to unseen attackers	Adopts anomaly detection (One-Class SVM, Isolation Forest, Autoencoders) trained on legitimate-user data only
Single anomaly-detection method per study	No direct, same-conditions comparison of leading one-class approaches for this task	Comparatively evaluates all three methods under the same multimodal feature set
Binary accept/reject decisions	Cannot express graded risk or support partial responses (e.g., step-up authentication)	Produces a continuously updated dynamic trust score with tiered responses
Short, lab-collected evaluation windows	Behavioral drift over realistic usage periods is rarely captured	Positions drift-aware, continuously updating trust scoring as a core design goal, not an afterthought
Generic device-unlock/session-security framing	Cost asymmetry of financial fraud versus false alarms is not modelled	Frames trust-score thresholds explicitly around mobile-banking risk (continue / step-up / terminate)

Table 3: Research gap summary: existing trends, their limitations, and how the proposed project addresses them.

5 Conclusion and Future Scope

The literature surveyed here traces a single, connected line of reasoning rather than a scattered collection of unrelated papers. Static, entry-point authentication was recognised early as insufficient for an extended session, which motivated continuous authentication built on behavioral biometrics. Keystroke dynamics was the first modality to mature, moving from statistical digraph distances to ensemble and deep sequence models [1, 7, 16, 15]. As smartphones became the dominant computing platform, the sensing surface expanded to touch, swipe, accelerometer and gyroscope data, and single-modality systems were increasingly replaced by multimodal fusion designs that trade the vulnerability of any one channel for the combined robustness of several [12, 13, 17, 11]. On the classification side, the field's dependence on labelled impostor data was progressively relaxed through one-class and autoencoder-based anomaly detection, which learns only what legitimate behaviour looks like [6, 10, 3]. Despite these advances, today's research still struggles with behavioral drift, small or short-duration datasets, unresolved usability and energy trade-offs, privacy concerns around continuous sensing, fusion strategies that are fixed rather than adaptive, and a still-thin body of adversarial robustness evaluation [9]. This combination of unresolved challenges is precisely what motivates a multimodal, comparatively-evaluated, dynamic-trust-score continuous authentication

system built specifically for the risk profile of mobile banking.

Looking forward, several directions raised implicitly or explicitly across the surveyed papers point to where this line of research is likely to go next. *Adaptive continuous authentication* that adjusts modality weighting in response to real-time signal quality, extending the static-fusion observations of Rayani and Changder [12] into an online mechanism. *Personalised behavioral models* that explicitly track and accommodate gradual behavioral drift rather than relying on a single enrolment profile, addressing the gap left open by every surveyed paper's short-duration evaluation. *Privacy-preserving authentication*, potentially through on-device inference or federated learning across users, so that the same passive sensing that enables frictionless security does not become a new privacy liability. *Energy-efficient mobile deployment*, given that several of the strongest-performing fusion architectures in this survey (transformer- and LSTM-based) are computationally heavier than the classical classifiers they outperform. And, most directly relevant to this project's own future work, *deployment and evaluation in genuine financial applications*, where the asymmetric cost of fraud versus false alarms – not just raw accuracy, FAR, or EER – should ultimately govern how a dynamic trust score is tuned and validated.

Beyond these directions, future work may also explore context-aware authentication that incorporates application usage patterns, adaptive threshold tuning based on individual behavior, and privacy-preserving collaborative learning across institutions. Such developments could enable continuous authentication systems to become both more resilient to evolving attacks and more practical for large-scale deployment in financial ecosystems.

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